

AGENTPROF: Semantic Profiling for AI Agents

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Abstract

AI agents orchestrate multi-step activities across users, tools, and system resources. To improve agent quality, safety, and cost efficiency, developers need to determine where failures concentrate, which workflows trigger unsafe effects, and which task categories consume the most budget. Traditional systems profiling answers analogous questions by aggregating resource use over code paths rather than debugging individual executions, but existing agent observability tools organize traces using run-local structure or application-supplied fields. Agent responsibility lies in semantic categories such as task intent and workflow phase rather than code paths, but trajectories lack stable identifiers and function-call hierarchies for automatic profiling attribution. Agent observability needs profiling, not only debugging. Our *semantic operation stack model* adapts profiling to agent trajectories by representing agent activities as uniform *operations* and replacing runtime stacks with *operation stacks* that attribute the same data hierarchically. AGENTPROF implements this model as a profiler with pluggable algorithms for *intent attribution* and *stack construction*, compiling local agent trajectories into pprof-compatible profiles. Across real and public trajectories, semantic profiles separate resource use by responsible semantic category and concentrate annotated problems. Out-of-fold evaluation of a pre-specified supervised boundary predictor on 287 session-held-out OSWorld-Human task instances yields 0.739 F1 versus 0.645 for the strongest simple control and 0.816 operation-weighted B³ partition F1 versus 0.678. AGENTPROF processes 27,765 operations in 1.17 s.

Introduction

AI agents (Yang et al. 2024; OpenAI 2025; Anthropic 2025; Xie et al. 2024) orchestrate multi-step activities across a high-level intent layer (prompts, LLM calls, and tool invocations) and a low-level system-effects layer (process spawns and file and network I/O). As agents evolve from short scripts into complex systems, production teams accumulate many trajectories over weeks or months.

To improve agent quality, safety, and cost efficiency, developers need to analyze operational behavior across trajectories and interactions. Which categories of tasks consume

the most token budget? Where do failures concentrate across workflows? Which behavioral patterns trigger unsafe system effects? At scale, manual trajectory inspection is slow and expensive (Lù et al. 2025), while per-trajectory LLM judging requires a separate evaluator pass for every run (Mazaheri and Mazaheri 2026; Zheng et al. 2023). Profiling provides the aggregation method these questions require. Traditional systems profilers attribute resource consumption to responsible entities rather than debug or trace individual executions.

Existing agent tools provide rich tracing, metadata aggregation, and population dashboards (LangChain 2024; Langfuse 2024; Arize AI 2024a; OpenTelemetry 2024; Datadog 2025; Laminar 2025). Yet these tools do not derive recurring semantic responsibility from heterogeneous histories or propagate it to downstream system effects as alternative profiler hierarchies.

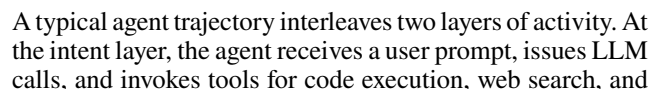
Adapting profiling to agent trajectories creates two challenges (§). Traditional profilers use stable function names to identify responsible code paths and runtime call nesting to provide the attribution hierarchy. Agent responsibility instead lies in semantic categories such as task intent and tool operation. Agent trajectories lack both properties. Natural-language prompts provide no stable identifiers for shared intent. Agent events also do not nest like function calls. A prompt sequence may represent one or multiple tasks, and one task may belong to a larger task.

Agent observability needs profiling, not only debugging. Despite these differences, the fundamental profiling methodology transfers to agent trajectories by projecting resources onto responsible entities, assigning stable tags, and attributing resources hierarchically. We propose a *semantic operation stack model* with two components. *Operations* are uniform records with string fields and additive measures that represent all agent activities (prompts, LLM calls, tool invocations, and system effects) without type-specific objects. *Operation stacks* provide hierarchical attribution, replacing the runtime call stack. Different field choices change the attribution granularity. The same operations can be attributed by task category, execution phase, session, or action type (Figure 1).

We implement this model in AGENTPROF, an offline profiler that compiles local agent trajectories into pprof-compatible profiles (Figure 2). Two pluggable mechanisms instantiate the model. *Intent attribution* derives stable, low-

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file editing. Each tool invocation triggers lower-layer system effects, including process spawns, file reads and writes, and network requests. A single trajectory may contain hundreds of such intent-effect cycles, and a production team accumulates many trajectories over time.

System Profiling

Profiling aggregates measurements to attribute resource consumption to responsible entities, in contrast to single-execution debugging and tracing. In traditional software, the profiling pipeline uses three steps: (1) a profiler periodically samples execution state, (2) it attaches each sample to the current call stack, and (3) it merges samples with identical stacks. Flame graphs (Gregg 2011) and pprof (Google 2024b) call graphs visualize aggregate cost as width or node size. Perfetto (Google 2024a) generalizes this with SQL-based analysis and derived events. These tools commonly begin with stable code identifiers and runtime stacks. Pprof can also aggregate labels and promote `tagroot/tagleaf` values to pseudo-frames. Agent trajectories instead must first obtain stable semantic responsibility fields and connect them to downstream effects before such profile views exist.

Challenges for Agent Profiling

Existing tools (LangChain 2024; Langfuse 2024; Arize AI 2024a; OpenTelemetry 2024) record agent events as per-execution span trees that support single-run debugging. AgentSight (Zheng et al. 2025) bridges intent and system-effect layers by correlating eBPF-captured system events with intercepted LLM traffic, but the resulting recordings still lack aggregate profiling.

Adapting profiling to agent trajectories faces two core challenges. The first is that the responsible entities differ. In traditional software, profiling attributes cost to code paths, but in agent trajectories the relevant entities are semantic categories such as task intent and workflow phase. Existing standards (OpenTelemetry 2024; Arize AI 2024b) represent application-supplied attributes in spans, but natural-language histories do not automatically provide stable responsibility categories because prompts or tool invocations expressing the same intent may use different strings.

The second challenge is that agent trajectories have no runtime call hierarchy that automatically supplies semantic attribution. In CPU profiling, the call stack determines attribution at every level. For example, `malloc`’s cost belongs simultaneously to `parse`, `process`, and `main`. A prompt sequence may constitute one task or several. A task may also belong to a larger workflow, yet no runtime mechanism marks these boundaries or determines the appropriate attribution granularity. We quantify this in RQ1 (§).

Design Requirements for Agent Profiling

Traditional profiling relies on three mechanisms that the call stack provides automatically. Agent profiling needs all three but must achieve them without a call stack:

D1: Cross-layer resource projection. In traditional profiling, each sample is taken inside a call context, so resource consumption is automatically attributed to the code path

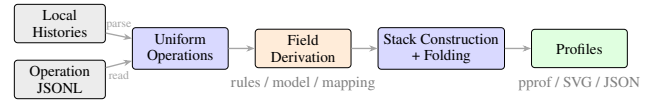


Figure 2: Data-flow and algorithm pipeline. Local histories enter through `agent-session` parsing, while public datasets enter as operation JSONL. Both yield uniform operations before field derivation, stack construction, folding, and profile export.

that caused it. Agent activities span an intent layer of prompts and LLM calls and a system-effects layer of file I/O and process spawns. The profiler must connect system-layer resource consumption to intent-layer semantic categories.

D2: Stable tags for folding. In traditional profiling, function names are stable identifiers that enable folding identical stacks. Agent prompts are natural language, so the profiler must derive short, stable, repeatable tags before folding.

D3: Hierarchical attribution. In traditional profiling, the runtime call stack provides the attribution hierarchy. Function calls nest, and folding identical stacks produces the tree that profiles visualize. Agent span trees do not encode semantic responsibility through function-call nesting, so the profiler must construct attribution hierarchies from the data.

Design

Cross-layer resource projection, stable tags, and hierarchical attribution drive the design. Operations (§) satisfy D1 by representing intent and system-effect activities in a single record with tag inheritance, so intent-layer tags propagate to the system effects they trigger. *Intent attribution* (§) satisfies D2 by deriving stable tags from natural-language fields. *Operation stacks* (§) satisfy D3. They replace the runtime call stack with query-time field projections that attribute resources at multiple granularities, allowing the same data to support multiple views without rebuilding the input.

The profiling pipeline has four stages: (1) parse raw trajectories into operations, (2) enrich operations via intent attribution or mapping rules, (3) project each operation onto a stack frame sequence chosen at query time, and (4) fold operations with identical stacks, summing their weights.

Semantic Operation Stack Model

Because agent activities span both the intent and system-effect layers described in §, a profiler should not distinguish between them in the data representation. AGENTPROF therefore represents every activity as an operation, a weighted record with string fields and additive measures. Each operation carries string fields (`project`, `agent`, `session`, `prompt_tag`, `kind`, `model`, `path`, `domain`, `status`, ...) and additive measures (token count, duration, event count). A prompt, an LLM call, a tool invocation, a file read, a GUI click, and a process event are all operations with the same schema, distinguished only by which fields

carry values. When a tool invocation triggers system effects, ingestion propagates its semantic fields to the resulting operations, allowing downstream resource weights to fold under the responsible intent.

An operation stack provides hierarchical attribution for agent trajectories, replacing the runtime call stack. In CPU profiling, the call stack determines which code paths share responsibility for a resource cost. An operation stack serves the same role. An ordered list of fields $[f_1, f_2, \dots, f_k]$ projects each operation o to a frame sequence $\langle o.f_1, o.f_2, \dots, o.f_k \rangle$, and operations with identical sequences are merged, summing their weights. Unlike a call stack, the fields are chosen at query time, not determined by execution. The same operations can therefore support a flat task summary, a per-session tree, a semantic phase/action hierarchy, or a trajectory organized by the dataset’s own annotations. Changing the fields changes attribution without changing the data. Sessions and spans are optional fields, not hierarchy levels. The same data supports debugging views with session and aggregate profiling views without that field. A view is a triple (φ, σ, w) . The predicate φ selects participating operations, the stack function $\sigma = [f_1, \dots, f_k]$ maps each operation to its frame sequence, and the weight function w assigns a nonnegative measure. Four built-in views cover common questions: (1) `tokens` weights LLM calls by token count, (2) `time` weights timed operations by duration, (3) `files` weights file operations by event count, and (4) `network` weights network operations by event count.

Algorithms

Intent attribution. Intent attribution maps natural-language prompts to short, stable, repeatable tags, giving agent trajectories the stable identifiers that traditional profiling gets for free from function names (§). The framework supports three pluggable backends. Explicit rules provide repeatable user control, a local model tags natural-language fields, and unsupervised grouping helps discover candidate categories for later rule authoring. Structured trajectories use the same interface to derive fields from existing action and quality attributes. All backends emit ordinary operation fields, keeping the stack model independent of how a tag was obtained.

Stack construction. Stack construction builds attribution hierarchies from available fields. When users supply an ordered list of fields, the projection is direct. When no field list is supplied, AGENTPROF induces operation stacks automatically. It detects boundaries from visible differences between adjacent operations and uses them to form variable-depth groups, while a configurable depth cap controls stack granularity. This mode produces coarser groups than hand-specified stacks but requires no domain knowledge, making it useful for initial exploration (§).

Implementation

AGENTPROF implements the semantic operation stack model as an offline Rust CLI (~9.8K LOC, Figure 2).

Input reconstruction. AGENTPROF reads Codex and Claude Code local JSONL history files and AgentSight (Zheng et al. 2025) recordings of system effects, then reconstructs prompts, LLM calls, tool invocations, and their triggered file and network effects.

Field derivation and boundaries. Regex rules are the production default. Each rule has the form `KIND:TAG=REGEX`, and rules use first-match semantics. Agents can refine the rules through repeated AGENTPROF runs until the unmatched rate falls below 5%, typically after 5–10 rounds. A local LLM tagger, such as a quantized 3B-parameter model through `llama.cpp` (Gerganov and contributors 2026), generates a single-word tag per prompt with grammar-constrained decoding and caching. TF-IDF (Salton and Buckley 1988) and K-Means (MacQueen 1967) discover candidate prompt groups without predefined rules, while mapping rules derive fields for structured trajectories, for example `phase:navigate=type:(click|scroll|key)`.

Boundary construction. Automatic stack construction scores adjacent operations using TF-IDF cosine shifts, structured-field changes, and group consistency before placing boundaries.

Profile export. Output formats are `pprof` protobuf for `go tool pprof -http`, folded stacks for `flamegraph.pl`, standalone SVG flame graphs, and JSON.

Evaluation

Research questions. The evaluation answers four fixed paper-level questions: **RQ1:** Does semantic profiling improve resource attribution? **RQ2:** Does profiler output correspond to real problems? **RQ3:** How accurate are the tags? **RQ4:** What is the profiling cost? Each following subsection states its protocol, reports the corresponding evidence, and gives the strongest answer supported by that evidence.

We use three classes of data. The real-agent dataset comprises 325 readable trajectories (198 Codex, 50 Claude, and 77 Claude subagent) and 183,714 system observations from a multi-month development project. The public annotated dataset comprises 15 real agent or human execution families converted by mapping rules into 47,590 operations. These cover web agents (WebLINX (Lù, Kasner, and Reddy 2024), Mind2Web (Deng et al. 2023), AgentTrek (Xu et al. 2025), WebShop (Yao et al. 2022)), API agents (API-Bank (Li et al. 2023), ToolBench (Qin et al. 2024), tau-bench (Yao et al. 2025)), coding agents (SWE-agent (Yang et al. 2024)), and mobile/GUI agents (AndroidCtrl (Li et al. 2024), GUI-Odyssey (Lu et al. 2025), ScaleCUA (Liu et al. 2026b)). Four mapped families provide structured quality or boundary annotations, including AgentRewardBench (Lù et al. 2025), SATraj-OS (Chen et al. 2026b), AgentNet (Wang et al. 2025), and OSWorld-Human (Abhyankar, Qi, and Zhang 2025). AgentProcessBench, HINTBench, and TraceElephant form the public problem-localization set and provide the independent targets for RQ2. Of HINTBench’s reported 629 trajectories, the current official test snapshot enumerates 536. We

use all 536 after selecting the field order on the separate 80-trajectory validation snapshot.

RQ1: Resource Attribution

RQ1 asks whether semantic profiling improves resource attribution beyond what traditional tag-free tools provide. We run a semantic-axis ablation on 325 real Codex/Claude trajectories (183,714 system observations). The prompt tags were produced by the local 3B-model backend described in § and are treated here as declared categories, not as an independent correctness oracle. We progressively add four tag-axis configurations: (1) no tags, (2) session only, (3) prompt only, and (4) both. We then measure mixed-weight percentage, defined as the fraction of system-effect cost in groups containing observations from multiple task categories. A lower mixed-weight percentage means fewer effects from different prompt-tag categories remain in the same responsibility group. The residual percentage measures weight in groups where the majority category accounts for less than 90% of the group, capturing partial mixing.

Figure 3 shows the result. Without semantic tags, nearly all system-effect cost falls into mixed groups. For example, a flat `cargo test counter` shows 2,903 executions but cannot distinguish whether they came from `review`, `debug`, `refactor`, or `test` prompts. The prompt tag is the decisive axis because it reduces both mixed weight and residual by more than half and nearly doubles the number of unique stacks. It separates observations that share the same system call but originate from different task intents. The session tag alone barely helps because trajectories typically contain multiple intent phases.

A tag-permutation test (1,000 shuffles, $p = 0.001$) confirms the separation is not random. The RQ1 experiment measures attribution conditional on the declared prompt tags. The independent RQ3 experiment evaluates group-boundary accuracy. These results show that prompt-level semantic fields materially refine responsibility views beyond tag-free aggregation.

Beyond tag separation, five field selections fold the same 13,265 annotated operations into 9 dataset, 57 phase, 226 semantic, 455 action, or 3,757 per-session groups. Unlike a single-tag `GROUP BY`, these depths answer task-budget, action-duplication, and session-detail questions. The same model covers all 15 trajectory families without type-specific objects.

The `time` and `tokens` views of the same 325 trajectories share only 7/10 top prompt categories (Spearman $\rho = 0.623$), and `network`-tagged prompts rank 8th by time but 93rd by tokens. Thus, multiple weights expose bottlenecks that one metric would miss.

Without user-specified fields, automatic induction on six annotation tasks produces a median of 12 groups, compared with 285 for the per-session view and one for the flat view, and yields variable-depth stacks on 4/6 tasks. We use independent problem annotations to score each group’s problem density. AP measures ranking quality, and inspection work measures the operations traversed in that ranking. Automatic induction achieves a median AP of 0.276 versus 0.312 for hand-specified stacks, while reducing inspection work to 65.3% of

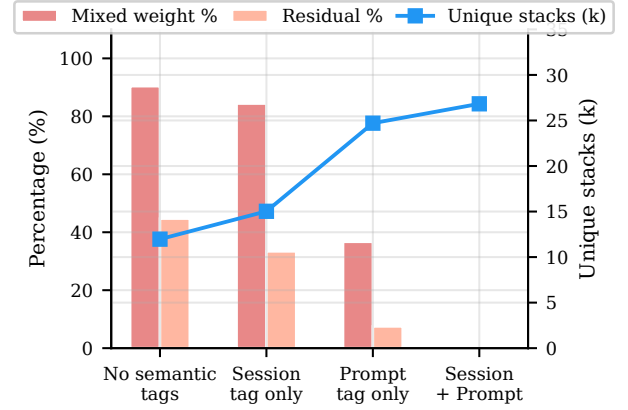


Figure 3: RQ1 semantic-axis ablation on 183,714 system observations from 325 real trajectories. Bars show mixed weight and residual percentages on the left axis, and the line shows unique stacks on the right axis. Adding prompt tags reduces mixed weight from 90.4% to 36.7% while increasing unique stacks from 12k to 25k. Session tags alone barely help (84.4%).

flat summaries without user configuration.

Figure 1 shows the resulting flame graphs for the `tokens` and `files` views, illustrating that changing only the stack fields and weight function produces different aggregates from the same operations. Together, these results answer RQ1. Semantic profiling improves resource attribution by separating mixed effects and providing multi-resolution, multi-weight, and automatically induced responsibility views.

RQ2: Problem Correspondence

RQ2 asks whether profiler output corresponds to real problems by testing whether a target-blind profile concentrates independently annotated failures or risky steps into groups that a developer can inspect early. We evaluate complete public workloads from AgentProcessBench (Fan et al. 2026), HINTBench (Wang et al. 2026b), and TraceElephant (Chen et al. 2026a). HINTBench development labels select among 24 fixed field orders. Its reported test labels and all AgentProcessBench and TraceElephant targets remain held out from tag, stack, and ranking construction. Only the final scorer reads them to compute inspection work or, for AgentProcessBench, operation-weighted average precision (AP) within each family and its equal-family macro average. AgentProcessBench uses 20 released judge votes as its step signal and ranks groups by mean risk. For HINTBench and TraceElephant, released localization predictions provide the step signal, and groups are ranked by the maximum 95% Wilson lower bound on hit density over their profile prefixes.

AgentProcessBench contains 1,000 trajectories from BFCL, GAIA, HotpotQA, and τ -bench with consensus human labels for all 8,509 assistant steps. Using the released target-blind risk signal, the semantic operation stack raises equal-family macro AP from 0.556 to 0.588. The paired gain is 0.0315 with a 95% interval of [0.0151, 0.0535].

Table 1: Target-blind RQ2 localization across public workloads. Work is the fraction of atomic steps inspected at the stated macro recall.

Workload	Steps	AGENTPROF	Raw action
AgentProcess	8,509	AP 0.588	AP 0.556
HINT	12,877	Work@80 41.57%	46.29%
TraceElephant	5,960	Work@50 19.55%	46.64%

Within each raw-action group, a permutation test shuffles semantic assignments while preserving subgroup sizes. Its $p = 0.00995$ shows that the gain comes from the semantic assignment rather than merely creating more groups. At a 30% inspection budget, recall rises from 35.85% to 43.52%.

The independent workloads show the same practical effect in different inspection regimes. On all 536 released HINT-Bench test trajectories, AGENTPROF reaches 80% macro recall after 41.57% of the steps, compared with 46.29% for raw action and 57.93% for the source-native order. On all 220 TraceElephant real failed executions, AGENTPROF reaches 50% macro decisive-step recall after inspecting 19.55% of the steps, versus 46.64% for raw action. At a 20% budget, AGENTPROF recovers 52.57% versus 23.79%.

These results answer RQ2 positively. Operation stacks expose recurring groups that are enriched for independently labeled problems and surface them earlier than raw-action inspection across three public workloads.

RQ3: Tag Accuracy

RQ3 asks how accurately categorical operation fields recover independent annotations. RQ3 covers task, phase, and action labels produced by intent attribution or mappings, as well as group boundaries used to construct the hierarchy. The present experiment evaluates group boundaries rather than intent labels.

We hypothesize that pre-specified target-blind taggers or mappings recover accurate, stable task, phase, and action labels and group boundaries on unseen agent and task families. Each component is scored against its corresponding independent annotations and then profiled with the original additive weights.

We test all 287 OSWorld-Human task-instance sessions with complete group annotations, covering 3,978 operations, 3,691 adjacent pairs, and 2,042 groups. Before evaluation, we fixed the supervised Bernoulli Naive Bayes predictor’s model family, nine input fields, adjacent-pair feature construction, and training procedure. Each session-blocked fold fits parameters and a threshold only on training sessions, then predicts every held-out session once. The predicted boundaries define an ordinary group field that AGENTPROF projects and folds through its released profile-construction path. This test covers supervised boundary prediction and field integration, not the TF-IDF automatic backend in §. The three simple controls place a boundary after every operation, whenever the visible action field changes, or whenever the visible phase field changes. We report boundary precision, recall, and F1, plus operation-weighted B^3 partition F1 (Bagga and Baldwin

Table 2: Session-held-out RQ3 boundary fidelity on OSWorld-Human. B^3 measures agreement between predicted and human operation partitions.

Method	Prec.	Recall	Boundary F1	B^3 F1
Supervised predictor	0.700	0.782	0.739	0.816
Always boundary	0.476	1.000	0.645	0.678
Action change	0.386	0.626	0.477	0.659
Phase change	0.441	0.268	0.334	0.665

Table 3: RQ4 profile-construction cost. Times are three-run medians, and RSS is the largest observed semantic-profile peak.

Workload	Ops	Sem. (s)	Raw (s)	RSS (MiB)
AgentReward	729	0.04	0.03	19.0
Satraj	4,285	0.17	0.14	73.5
OSWorld	6,010	0.25	0.21	108.4
AgentNet	16,741	0.71	0.58	278.4
Union	27,765	1.17	0.99	464.5

1998), whose per-operation predicted–human group overlap captures merging and fragmentation.

The supervised predictor reaches 0.739 boundary F1, 0.094 above the strongest simple control, and has the highest boundary F1 in all five held-out folds. Its groups reach 0.816 B^3 F1 against the human partition, compared with 0.678 for the strongest control. Equivalently, $(1 - B^3 \text{ F1})$ falls from 0.322 to 0.184. AGENTPROF projects all 3,978 operations onto the predicted group field, producing 2,249 stacks while preserving the total weight of all 3,978 operations. This result supports the group-boundary component of RQ3 on session-held-out OSWorld-Human task instances.

RQ4: Profiling Cost

RQ4 asks what it costs to construct profiles after a session ends. Because AGENTPROF constructs profiles after execution, profile construction adds no online overhead. We measure its post-session time and resources, excluding trajectory capture. We measure `agentpprof 0.2.37` on four complete existing public operation workloads and their exact union. For each input, we run a fixed semantic hierarchy and an identical-input raw-action cost control three times. Both execute the full parsing, stack-construction, folding, and serialization path. The measurements run on a 24-core Intel Core Ultra 9 285K machine with 125 GiB RAM and Linux 6.15.11.

Both time curves are monotonic across the five measured input sizes. The semantic curve has a descriptive slope of 0.0422 ms per operation ($R^2 = 0.9997$), reaching 23,731 operations/s on the union. At that largest input, semantic construction adds 180 ms (18.2%) and 6.0 MiB maximum RSS (1.3%) over raw action.

Field derivation is cacheable. On matched inputs from eight real Codex sessions, the predecessor AgentFlame used 60 local-model calls and 1.64 s. Cached reconstruction used no calls and 0.11 s with 76/76 hits. Thus, shared tag caching

avoids repeated field-derivation calls. This predecessor result is not timing for the current `agentpprof` binary. Together, these results answer RQ4. Profile construction is practical and predictable over the tested range, and cached derivation avoids repeated model work.

Scope and Limitations

The evaluation covers offline profiling on tested local trajectories and public workloads. It excludes live-agent overhead because `AGENTPROF` constructs profiles after execution.

RQ3 currently establishes group-boundary fidelity on session-held-out OSWorld-Human task instances. Matched task, phase, and action annotations and the corresponding pre-specified taggers or mappings across additional agent families remain the evidence required to complete the broader fixed RQ3 hypothesis.

Related Work

Agent observability tools (LangChain 2024; Langfuse 2024; Arize AI 2024a; OpenTelemetry 2024; Arize AI 2024b; Dong, Lu, and Zhu 2024; Balusu 2026; Wang et al. 2026c; Datadog 2025; Laminar 2025) combine per-execution traces with tag, metadata, signal, and metric aggregation. `AGENTPROF` instead derives recurring semantic fields across heterogeneous histories, propagates them to additive cross-layer effects, and constructs query-time profile hierarchies.

Traditional profilers aggregate stack samples and can incorporate supplied labels into profile views (Google 2024b; Gregg 2011). Trace systems such as Perfetto support flexible event queries (Google 2024a). `AGENTPROF` derives these fields from agent histories, connects them to downstream effects, and constructs alternative hierarchies over the same operations.

Recent agent-diagnosis work spans step- or span-level localization (Barke et al. 2026; Wang et al. 2026a; Liu et al. 2026a), validation with root-cause analysis (Mulian et al. 2026), and diagnostic taxonomies and audit protocols (Mazaheri and Mazaheri 2026). `AGENTPROF` complements these methods with category-level aggregate profiling across trajectories, exposing recurring responsibility groups rather than only one execution’s fault location.

Conclusion

Agent observability needs profiling, not only debugging. `AGENTPROF` shows that the semantic operation stack model, with pluggable intent attribution and stack construction, brings hierarchical attribution to agent trajectories. Across 325 real trajectories, semantic fields separate effects left mixed by tag-free and per-session views, revealing distinct token, time, and system-effect responsibility profiles. Across `AgentProcessBench`, `HINTBench`, and `TraceElephant`, target-blind operation stacks improve annotated-problem concentration over raw-action inspection at the workloads’ reported operating points. On 287 session-held-out OSWorld-Human task instances, out-of-fold evaluation of the pre-specified supervised boundary predictor yields 0.739 F1 versus 0.645 for the strongest simple control and 0.816 operation-weighted B^3 F1 versus 0.678. On 27,765

public operations, `AGENTPROF` builds the complete semantic profile in 1.17 s with 464.5 MiB maximum RSS, adding only 18.2% time and 1.3% memory over raw action. Together, these results establish cross-trajectory semantic profiling as a complement to run-local debugging and tracing.

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